Bedfordview Primary School
TERM 4 TEST - MEMO

Name: ______________________________ Grade 6 : ___
Subject: **Natural Sciences & Technology**
Examiner: **Miss Z Kajee & Mrs N Suliman** Date: **November 2023**
Moderator: **Mr J Nash** Time Allocation: **1 ½ hours**

INSTRUCTIONS

1. An extra 10 minutes will be provided for reading.
2. Write neatly and legibly.
3. Answer in the spaces provided.
4. This paper consists of 12 pages.

Parent's signature: ______________________
Comments: ______________________
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	Section A /30				**Section B** /30		**Teacher comment:** ______	
Question	**1**	**2**	**3**	**4**	**5**	**6**	**TOTAL**	**%**
Allocated mark	**4**	**4**	**19**	**3**	**20**	**10**	**60**	
Learner's mark								
Moderated mark								

1	**2**	**3**	**4**	**5**	**6**	**7**
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

SECTION A – ENERGY AND CHANGE **[30 Marks]**

Question 1: Definitions (4)

Match the term in **Column A** with its description in **Column B.** Write the correct letter next to the question number in the space provided.

Column A	Column B
1.1 Hydro-electric power	A. The network of electricity cables that transport electricity from the power station to our homes.
1.2 Non-renewable energy	B. Energy resources that can be produced continuously such as wind and hydroelectric energy.
1.3 Renewable energy	C. Energy sources that cannot be replaced once used such as coal; oil; gas and nuclear fuels.
1.4 National grid	D. Electricity produced using the energy from fast-flowing water.

1.1 D√ 1.2 C√ 1.3 B√ 1.4 A√

Question 2: True or False (4)

State whether the following statements are **TRUE** or **FALSE**. **If false , correct the statement to make it true.**

2.1 Materials that resist the flow of electricity are called conductors.

False. √

Materials that resist the flow of electricity are called insulators OR Materials that allow for the flow of electricity are called conductors. √

2.2. Fossil fuels are natural fuels which forms from dead animals and plants over millions of years.

True √

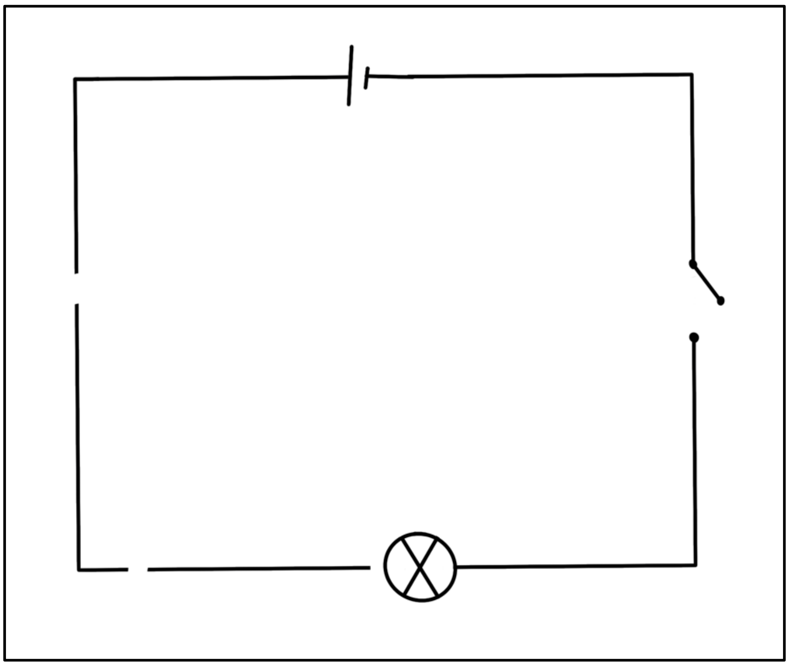
2.3. A battery has two terminals; a positive and a negative terminal.

True √

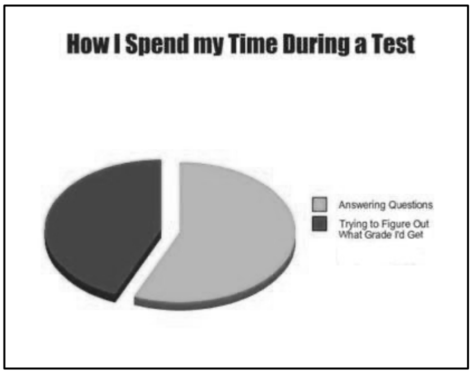
Question 3: Electric Circuits and Systems (19)

3.1 Study the electric circuit diagram below and answer the questions that follow.

3.1.1 Look at the circuit diagram above. **The bulb does not light up for 4 reasons**. **Draw a circle** around the parts of the circuit that prevents the light bulb from lighting up. (4)



3.1.2 Redraw the above circuit diagram **if the light bulb had to shine**. Ensure your diagram has: ***A battery of 2 cells, wire, a light bulb, and a switch. You should indicate the positive and negative terminal on the battery and show the flow of current.*** (5)



You can do it! Keep doing your best!

3.2. Read and study the following article and answer the questions that follow.

SOUTH AFRICA

Fearing for their lives, Eskom workers abandon plan to cut illegal connections

15 June 2023 - 18:24

Phathu Luvhengo

Eskom technicians abandoned part of their operation to remove illegal connections in Marlboro industrial area on Thursday for safety reasons. People living illegally in an abandoned factory had allegedly connected electricity illegally and threatened to burn down nearby factories if Eskom cut the cables. The utility's technicians initially disconnected residents in Stjwetla informal settlement in Alexandra, Johannesburg, who had built their homes under Eskom pylons. They had illegally connected cables to a substation in Marlboro Gardens.

One of the residents, Charlotte Motseo from Limpopo, complained that the settlement was not safe without electricity. She insisted Eskom should allow them to connect. She said if she had a job she would have probably stayed in a better place and there would not be a need for residents to illegally connect electricity. She said she would boycott the next year's elections. "How will we vote if they're doing this to us? They could just leave us or electrify our area so that we will be safe. Now they have disconnected us and there is a lot of crime here," she said.

The utility disconnected the area where residents had built shelters under the Eskom 88kV power lines. Technicians disconnected the exposed live wires from the informal settlement.

Just a few metres away from the settlement is the industrial area, where one of the businessmen who had been operating for more than seven years claimed residents had allegedly threatened to burn down his factory. He said people living in one of the abandoned factories connected a cable to a substation that connects to his factory, which manufactures water bottles. "If that cable comes down now, tomorrow morning you will find this factory burnt to the ground," he said. He pleaded with Eskom technicians not to disconnect the cables. He said residents told him several times that if they don't have electricity the entire neighbourhood should be disconnected as well.

For fear for their safety and citing limited law enforcement agencies and a few private security guards who had accompanied them, Eskom technicians eventually abandoned their operation.
Eskom senior manager for maintenance and operations in Gauteng, Mashangu Shivambu, said the utility suffered an annual loss of about R7bn in the province due to non-technical issues usually caused by illegal connections.

He said when Eskom disconnected the wires, residents usually reconnected them. "Unfortunately, it is a repeated cycle. We remove it today, and the next day they reinstall — but we can't just fold our arms. "Whatever happens here affects everyone. These illegal networks result in customers being constantly off because of the faults that are generated from these networks," he said. He said it was not only dangerous but also illegal for residents to build homes under Eskom's pylons or high-voltage power lines.

"If you do get a fault on a pylon network, that fault is going to go through those pylons directly to the ground and it becomes very dangerous. People can get killed even if they are not directly connected to the pylons," he said.

TimesLIVE https://www.timeslive.co.za/news/south-africa/2023-06-15-fearing-for-their-lives-eskom-workers-abandon-plan-to-cut-illegal-connections/

3.2.1 What was the **reason** for the **operation** performed by the **Eskom workers**? (1)

The Eskom workers had to go on their operation due to the illegal connection activities. √

3.2.2 **Discuss** any **2 dangers** which may result because of the actions of the residents. (2x ½ =1)

It is dangerous. √ ½

It can cause an overload and power outages. √ ½

3.2.3 **Explain** why people need to **pay** for electricity. (1)

People need to pay for electricity as the cost to produce electricity is very high and the cost of infrastructure is high. The higher the demand for electricity the higher the cost of electricity. (infrastructure = key word in answer) √

3.2.4 **Why** did the Eskom technicians **abandon** their operation? (1)

The Eskom technicians abandoned their operation as it became too dangerous to cut off illegal connections – for their safety. √

3.3. <u>**Cost of Electricity:**</u>

The grade 6 learners at Bedfordview Primary School investigated which electrical appliances uses more electricity to run than others. They looked at the back or bottom of each appliance and recorded the information in the table as shown below.

Study the table and answer the questions that follow.

Appliance Number:	Name of Appliance:	Power Rating in Watt (W):
1.	LED light bulb	20W
2.	Fridge	120W
3.	Heater	100W
4.	Incandescent lightbulb	40W

3.3.1 **Draw a bar graph** showing the power rating in Watts (W) for each appliance to help the grade 6 learners identify which appliance uses more electricity. (6)

Remember to **write all your numbers in blue pen**. Your graph is to be **drawn in pencil**.

Remember to label both the x and y – axis. You must give your graph a suitable heading.

Your graph will be marked accordingly:

- **Heading (1)**
- **Y axis heading (1)**
- **Axis scale (2 x ½ = 1)**
- **X axis heading (1)**
- **Bars (4 x ½ = 2)**

Question 4: Renewable and Non-renewable Energy Sources **(3)**

4.1 The table below shows how coal is formed over millions of years. The steps have been mixed up. Arrange the steps in the correct order by writing only the missing numbers on the right.

(3)

Incorrect order	Correct order
Layers of coal are formed.	**4**
When the plants died, they were covered under layers of sediments over millions of years, after sinking to the bottom of swamps.	**2 √**
Millions of years ago there were large forests in swampy areas.	**1 √**
Pressure and heat compacted the sediment layers.	**3 √**

TOTAL SECTION A: 30

Turn over for Section B

SECTION B: PLANET EARTH AND BEYOND **[30 Marks]**

Question 5: Planets and our Moon **(20)**

5.1. Using the clues, unscramble the following terminology.

5.1.1. A system consisting of many stars, together with dust and gas.

lxaagy: galaxy √ (1)

5.1.2. The shape in which the planets revolve around the Sun.

pteliiillca: elliptical √ (1)

5.1.3. A human-made or natural object that revolves around a planet.

teasiltle: satellite √ (1)

SPELLING is very important. If any letter is missing, the answer is incorrect.

5.2. Complete the following table based on the amount of time it takes for the planets to revolve around the Sun.

Planet:	**Number of Days or Years:**
Jupiter	5.2.1 **12 years** √ (1)
Neptune	5.2.2 **165 years** √ (1)
5.2.3 **Earth** √ (1)	365 ¼ days

5.3. Look at the picture of the moon below and answer the questions that follow.

5.3.1. **Why** is the moon **cold**?

The moon is cold because it does not give out its own heat √ and because it does not have its own atmosphere. √ (2)

5.3.2. **Differentiate** between the **lighter areas** on the moon and the **darker areas** on the moon.

The light areas on the Moon are mountains. √

The dark areas on the Moon are flat plains. √ (2)

5.3.3. **Explain** why we see different **phases on the Moon** over a period of a month. Begin your answer by defining "**phases of the Moon**."

Phases of the moon refers to the changing shape of the bright part of the Moon. √

The movement of the Moon around the Earth causes us to see the lit up parts of the Moon from different angles. √

The part of the Moon that can be seen is the part that the Sun is shining onto. √ (3)

5.4. The Sun is a star at the centre of the Solar System. The planets and all the other objects in the Solar System revolve around the Sun in a pathway called an orbit. The Earth revolves around the Sun. The Earth spins on its own axis.

Using the PEEL method, write a paragraph **explaining the difference between a rotation of the Earth, the revolution of the Earth and how it affects our lives on Earth**.

Your paragraph should answer the following questions:

- What is an axis of the Earth? (1)

An axis is the imaginary line that runs from one pole to the opposite pole. √

- What is a rotation of the Earth? (1)

The rotation of the Earth is the act of one full spin of the Earth on its axis. √

- How does the rotation of the Earth affect our lives on Earth? (1)

The rotation of the Earth on its own axis allows us to experience night and day. √

- Explain how we are able to experience daytime and night-time. (2)

The side of the Earth that faces the Sun experiences daytime. √

The side of the Earth that faces away from the Sun experiences darkness and night-time. √

- What is the revolution of the Earth around the Sun? (1)

The revolution of the Earth is one complete movement along a pathway or an orbit around the Sun. √

- How does the revolution of the Earth around the Sun affect our lives on Earth? (1)

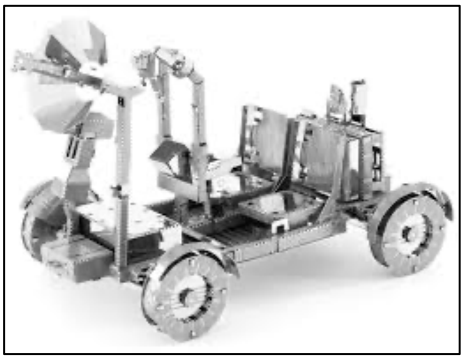
The revolution of the Earth around the Sun allows us to experience the 4 seasons. √(7)

<u>Question 6: Systems for looking into space</u> **(10)**

6.1. Complete the table below based on your knowledge and understanding of telescopes.

	Description/Function of the telescope	**Example**
Optical Telescope	**An optical telescope uses mirrors, lenses or a combination of both mirrors and lenses to catch light waves from objects in space. √** (1)	**The Hubble telescope √** (1)
Radio Telescope	**Radio telescopes use a dish to collect and concentrate radio waves from objects in space. A computer uses the information to draw a picture of the source of the waves. √** (1)	**MeerKAT array √** (1)
What is a telescope?		
A telescope is an instrument that MAGNIFIES distant objects. √ (1)		

6.2. You are recruited to work for an astronomer, who is planning a trip to the Moon. Before departing on your trip, you need to investigate the surface of the moon. **Explain how you could use a moon rover in order to investigate the surface of the moon.**



Answer the following questions to build your project on investigating the surface of the moon.

6.2.1. What is a moon rover?

A moon rover is a vehicle that is specifically developed to be driven over the surface of the Moon. √ (1)

6.2.2. Identify 3 design specifications of your moon rover.

- **Works in a low-gravity environment with rocks and craters and fine dust.**
- **Works in an environment with a wide temperature range.**
- **Fold up to fit inside the spacecraft easily.**
- **Be lightweight.**
- **Be able to carry two astronauts at a time as well as equipment and rock samples.**
- **Work in an environment with no air or water.**
- **Use an energy source other than petrol.**

(Any 3) √ √ √

(3)

6.2.3. Define the term "astronomer".

An astronomer is a scientist who studies objects in space. √ (1)

TOTAL SECTION B: 30

<u>TOTAL: 60</u>

-- END OF TEST---